

BCSE 3072 - Compiler Design
Digital Assignment 1

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: C1+TC1, C2+TC2

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1. Construct the predictive parser for the following grammar and show the behaviour of the parser in the string $(a, (a, a))$

$$S \rightarrow a \mid \uparrow \mid (T)$$

$$T \rightarrow T, S \mid S$$

2. Check whether the following is LL(1) grammar

$$S \rightarrow iEtS \mid iEtS_eS/a$$

$$E \rightarrow b$$

3. Construct Operator precedence parser for the following grammar and parse the string $id=id$

$$S \rightarrow L=R \mid R$$

$$L \rightarrow *R \mid id$$

$$R \rightarrow L$$

4. What is recursive descent parser? Construct a recursive descent parser for the following grammar and check whether the string $id*id+id\$$

$$E \rightarrow T+E \mid T$$

$$T \rightarrow V* \mid V$$

$$V \rightarrow id$$